Team Number: 89

M2 — Automatic Category Suggestions for Wikimedia Commons

Problem Statement 3 · Progress report

1. Current approach and changes since M1

The architecture is unchanged from M1 and fully implemented: a frozen CLIP ViT-B/32 encoder; four scorers (k-NN voting over categorised images, zero-shot category-name prototypes, a text-initialised multi-label head trained with asymmetric loss, and a metadata scorer using title/description embedding, word overlap and year agreement); logistic-regression fusion over the union of candidates; hierarchy post-processing; live index updates from editor feedback and new categories. Working: data crawler, training, evaluation harness, command-line predictor, and a web demo showing evidence images and accept/reject feedback. Changes since M1:

- **Hierarchy strength tuned on validation.** With a hand-set penalty, ancestor suppression lowered strict F1@5 (0.127 → 0.119) because some files *are* labelled with broad categories. The penalty is now chosen on the validation split (1.0 = off), so the rule is applied only as strongly as it helps.
- **Portable search backend.** On macOS, FAISS and PyTorch can deadlock (both bundle OpenMP); exact NumPy search is now the default there, with FAISS (flat/HNSW/IVF-PQ) used on Linux and at scale.

2. Experiments and results

Run 1 setup. 1,279 Commons files crawled through the MediaWiki API from 107 seed categories (objects, scenes, events, abstract concepts; 12 countries across Asia, Africa and Latin America) and their subcategories. Labels are the files' editor-assigned categories after removing hidden and maintenance categories. Temporal split by upload time: 898 train / 191 validation / 190 test; 0 near-duplicates found (perceptual hash + embedding cosine ≥ 0.95). Vocabulary: 4,876 categories, of which 159 have ≥ 3 training images (supervised head). k-NN tuned on validation: K = 50, τ = 0.1.

Table 1. Test set, fusion + hierarchy (sample-wise).

input	k	P@k	R@k	F1@k	hier. P@k
image + metadata	1	0.284	0.113	0.148	0.384
image + metadata	5	0.096	0.202	0.119	0.194
image + metadata	10	0.057	0.238	0.086	0.149
image only	1	0.247	0.113	0.142	0.353
image only	5	0.072	0.156	0.089	0.145
image only	10	0.040	0.178	0.061	0.098

mAP (macro over test categories) 0.174. Label coverage 0.335.

Table 2. Ablation, image + metadata.

ranking	F1@5	R@10
fusion (no hierarchy)	0.127	0.258
fusion + hierarchy (fixed 0.3)	0.119	0.238
k-NN only	0.084	0.171
metadata lexical only	0.082	0.129
supervised head only	0.077	0.164
metadata text only	0.066	0.157
zero-shot only	0.057	0.133

Table 3. Recall@5 by category frequency in the index.

bucket	categories	labels	macro R
unseen (0)	231	552	0.024
tail (1–4)	21	65	0.134
torso (5–19)	36	99	0.585

Table 4. By category type (k = 5).

type	labels	recall	precision
object	531	0.121	0.100
scene	68	0.176	0.090
event	28	0.214	0.087
abstract	89	0.101	0.084

Table 5. Robustness, F1@5 (share flagged low-quality).

degradation	flagged	+ metadata	image only
none	–	0.119	0.089
Gaussian blur r=2	1.00	0.115	0.077
Gaussian blur r=5	1.00	0.106	0.054
JPEG quality 10	0.28	0.117	0.081
downscale ×0.15	1.00	0.112	0.063

Efficiency (measured on the team's Mac, no server GPU): end-to-end per image p50 16.9 ms, p95 128.2 ms; scoring and ranking 0.61 ms/image; index of 1,089 images and 4,876 categories uses 11.7 MB in memory.

Findings. (1) **Vocabulary coverage is the main ceiling:** only 33.5% of test labels exist in the vocabulary, so recall cannot exceed 0.335; R@10 = 0.238 already recovers ≈71% of reachable labels. (2) **Fusion works:** +51% F1@5 over the best single scorer. (3) **Metadata helps** (+34% F1@5) and keeps accuracy stable under heavy degradation, where image-only accuracy drops by up to 40%. (4) **Examples per category matter most:** recall@5 is 0.585 for categories with 5–19 examples but 0.024 for unseen ones. (5) **Strict metrics under-count:** hierarchy-aware P@1 is 0.384 vs strict 0.284, so about one in seven top-1 "errors" is a parent or child of a true label. (6) Events score best on recall (year and title matching).

3. Progress on testing

- Evaluation harness complete with every metric planned in M1: P/R/F1@k, mAP, hierarchy-aware precision, frequency buckets, category types, resolution bands, synthetic degradations, latency and memory, plus per-scorer ablation.
- 59 automated tests: category filtering, splits, metrics, index search (FAISS and NumPy), hierarchy rules, the trained head, fusion, an offline end-to-end run (train → evaluate → predict → update) on synthetic data, and encoder API checks on tiny CLIP and SigLIP models.
- Continual learning verified: categories added at run time and accepted feedback are suggested immediately without retraining; rejected labels act as negative votes.

4. Plan for the remaining time

- **Run 2 (in progress):** ~6,000 files with more files per category, to move categories from the unseen/tail buckets into the torso, where recall is highest.
- **Raise label coverage:** add category names without images (subtrees of the seeds and siblings of known categories), so the zero-shot and lexical scorers can propose them; measure coverage and unseen-bucket recall.
- **Encoder comparison:** SigLIP-base vs CLIP ViT-B/32 on the same split; report the accuracy/latency trade-off.
- **Reviewed score:** manually check a sample of “wrong” suggestions to estimate how many are correct but unlabelled.
- **Deliverables:** demo video (web UI with evidence and live feedback), final report, slides, externally hosted index.